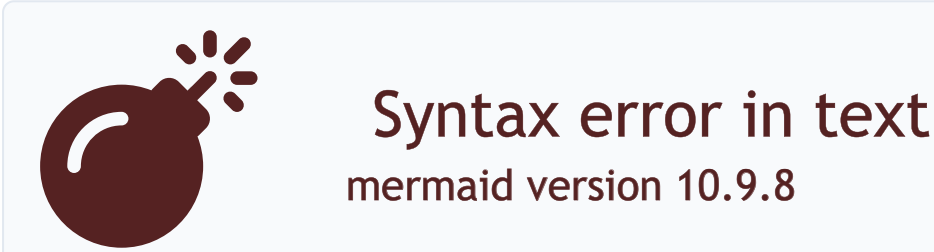
# Submodule 36.10: PyTorch Neuron Interactive Notebook (Hinglish)

## 1. Notebook ka Structure aur File Link

Yeh document lecture notebook ka complete cell-by-cell breakdown provide karta hai:

- **Jupyter Notebook File**: `10_lecture_code_Neuron.ipynb`
- **Framework**: PyTorch ( `torch` , `torch.nn` )




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## 2. Har Cell ka Explanation

### Cell 0: Import

```
import torch
import torch.nn as nn
```

PyTorch aur neural network module ko import kiya gaya.

### Cell 1: Inputs

```
torch.manual_seed(42)
inputs = torch.tensor([1.0, 2.0, 3.0])
```

3 features ka ek input tensor create kiya gaya.

### Cell 2: Single Neuron Layer

```
neuron = nn.Linear(in_features=3, out_features=1)
```

Neuron banaya gaya jisme 3 inputs aate hain aur 1 output nikalta hai.

### Cell 3 & 4: Forward Pass

```
output = neuron(inputs)
output
```

Inputs ko neuron ke andar pass karke output generate kiya gaya.

### Cell 5 & 6: Mathematical Proof

```
neuron.weight @ inputs + neuron.bias
neuron.bias
```

`@` operator matrix multiplication karta hai. Prove ho gaya ki PyTorch andar wahi $W \cdot X + B$ formula chala raha hai.